HASNAT MD. ABDULLAH

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Research Interest

- My primary research focuses to bridge the gap between AI and genetics to build a more sustainable beef industry. By integrating SNP genotypes with 3D computer vision, I develop scalable deep-learning tools that transform raw imagery into precise efficiency traits. In addition, my other research works cover VLM/LLM Alignment, Video Language Modeling, UAV / Sattelite imagery, Commercial Structure imagery and Disease Surveillance.

Education

PhD in Animal Sciences, University of Illinois Urbana-Champaign, Urbana, IL Aug. 2025 – Present
Masters in Computer Science, Texas A&M University, College Station, TX Jan. 2023 – July 2025
Bachelor of Science in Computer Science, BRAC University, Dhaka. Jan. 2017 – Dec 2020

Work / Research Experience

Research Assistant at Illinois Phenomics and Genomics Group August 2025 – Present

- Advisor: Dr. Tiago Bresolin
- Topic: High-throughput phenotyping of beef cattles using 3D vision technologies.

Summer 2025 Internship at AIISC Lab, University of South Carolina (Onsite) May 2025 – July 2025

- Advisors: Dr. Amit Sheth
- Topic: Encoder free Vision Language Model; building small language model (SLM) for custom and compact enterprise solutions.

Collaborator at IDRT Jan 2025 - Aug 2025

- PI: Dr. Jooho Kim
- Topic: Understanding the Physical Structure of Communities for Improved Disaster Impact Predictions.

Collaborator at Humanitarian Robotics and AI Laboratory Jan 2024 - Aug 2025

- PI: Professor Robin Murphy
- Topic: Building and Road Damage Assessment via CV/ML techniques from Georectified small uncrewed aerial systems (sUAS) Imagery.

GRA at TAMU Natural Language Processing Group Jan 2023 – Dec 2023

Publications

A spatiotemporal GNN for predicting New World Screwworm outbreak risk and informing sterile fly release.

- **First author**. Manuscript under review at *Journal of Animal Science*. 2026

Evaluating the Impact of Depth Image Encoding and Temporal Domain Shift on Cattle Body Segmentation.

- **First author**. Manuscript under review at *Computers and Electronics in Agriculture*, Elsevier. 2026

DETONATE: A Benchmark for Text-to-Image Alignment and Kernelized Direct Preference Optimization. 2026

- **First author**. Published in AAAI-26 Special Track on AI Alignment

CliME: Evaluating Multimodal Climate Discourse on Social Media and the Climate Alignment Quotient. 2025

- **First author**. Published in ACL2025-NLP4PI Workshop.

InstaLearn: Continual Knowledge Updates for Domain Adaptation. 2025

- **First author.** Accepted for the KDD'25 International Workshop on Resource-Efficient Learning for Knowledge Discovery (RelKD).

UALBench: The First Comprehensive Unusual Activity Localization Benchmark. 2024

- **First author.** Published in WACV'25.

Non-Uniform Spatial Alignment Errors in sUAS Imagery From Wide-Area Disasters. 2025

- **Co-author**. Published in IEEE RO-MAN 2025 : 34th IEEE International Conference on Robot and Human Interactive Communication.

Alignment Quality Index (AQI). 2025

- **Co-author**. Published in EMNLP 2025 Main Conference.

Mitigating Gender Bias via Fostering Exploratory Thinking in LLMs. 2025

- **Co-author**. Accepted in EMNLP 2025 Findings.

Ask Me Again Differently: GRAS for Measuring Bias in VLMs on Gender, Race, Age, and Skin Tone. 2026

- **Co-author**. Published in EACL 2026

Visual Counter Turing Test (VCT^2). 2024

- **Co-author.** Published in IJCNLP — AACL 2025.

SynthEnsemble: A Fusion of CNN, Vision Transformer, and Hybrid Models for Chest X-Ray Classification. 2023

- **Co-author**. Published in IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET) 2022.

A Comparative Analysis of Lumpy Skin Disease Prediction through ML approaches. 2022

- **Co-author**. Published in IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET) 2022.

Affective Social Anthropomorphic Intelligent System. 2023

- **Co-author**. Published in Multimedia Tools and Applications Journal, Springer Nature, 2023.

Honors and Awards

- Second Place Winner, Smart Livestock Track, Precision & Digital Agriculture Hackathon, Center for Digital Agriculture, University of Illinois Urbana-Champaign (2026).
- Outstanding Paper Award, 4th Workshop on NLP for Positive Impact, ACL 2025.
- Travel Grant Recipient ($1,000), Department of Computer Science and Engineering, Texas A&M University — IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025).
- Quality Journal Publication Award, BRAC University — for publication in *Multimedia Tools and Applications* (Q1 Journal, Vol. 82, Issue 23, 2023).
- Top 120 Startup Finalist, Student to Startup Program (Chapter 1), Innovation Design & Entrepreneurship Academy, Bangladesh (2019).
- Vice Chancellor's List, BRAC University: Spring 2018, Summer 2018, Fall 2019, Summer 2020.
- Dean's List, BRAC University: Fall 2017.

Reference

1. Dr. Tiago Bresolin

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2. Dr. Amit Sheth

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